

Ultrashort laser sintering of metal nanoparticles: A review

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ABSTRACT

The unique shape and size dependant properties of nanoparticles (NPs) are responsible for their emerging, extensive applications in everyday industry. The application of NP inks for printed flexible electronics requires an additional post processing step since the presence of additives and stabilizers in the NP based inks often results in inferior electrical properties of the printed patterns. Laser sintering is a well-established technique for past number of years and is extensively used with continuous wave (CW), long, and short pulsed lasers. The literature is enriched with conventional sintering methods however the field of ultrashort laser sintering is relatively under-reported compared to CW laser and nanosecond laser-based techniques. This study concisely focuses on the potential use of ultrashort laser pulse processing of NPs and NP based thin films. The review provides the recent advances on ultrashort lasers sintering with a key focus on femtosecond laser pulses.

1. Introduction

This review focuses on the laser post processing of NPs and NPs based thin films. Different processing routes involving sintering, annealing, crystallization, and welding are employed to process nanomaterials depending on the composition, structure, and processing requirements. Sintering is defined as the coalescence of particles together to form solid or porous mass without liquification [1]. Sintering technique is extensively used for NP inks to fuse them together by eliminating the solvent. This yields in enhanced electrical and structural properties required for printed electronics applications [2]. The improvement in electrical, optical, and structural properties of nanomaterials can be obtained by transforming an amorphous material to crystalline phase through a crystallization process [3]. Additionally, growth of crystal phases with temperature through an annealing process also results in good electrical and structural properties [4]. Laser driven welding of NPs is also performed to form larger nanostructures by joining similar and dissimilar NPs together [5]. However, the laser-material interaction is often a combination of more than one processes where the absorbed energy could result in crystallization: either through a sintering route or, through an annealing process depending on the laser and processing parameters [6,7].

Laser sintering of NPs is broadly adopted, and extensive studies are reported in literature in past two decades. Where the interest in CW and

nanosecond laser sintering has grown significantly, there is not much data reported for ultrashort pulsed laser sintering. This contribution is devoted to review the progresses made in ultrashort laser sintering using picosecond (ps) and femtosecond (fs) laser pulses for printed electronics. The coalescence of NPs in a printed layer results in improved electrical and mechanical adhesion when subjected to ultrashort laser pulses. The use of ultrashort pulses also facilitates the crystallization of NPs in a thin film due to laser annealing process. Starting with the basic building block of the printed electronics, we will first primarily discuss metal NPs, their properties, applications and how they are deposited. A concise literature survey of ultrashort laser sintering is provided in later sections with a key focus on femtosecond laser pulses.

2. NPs and NPs based thin films

When the size of a bulk material changes and is reduced to a scale of nanometres ($1 \text{ nm} = 10^{-9} \text{ m}$), it causes a dramatic change in the chemical and physical properties which remarkably vary from those of the bulk; following to a class of nanomaterials classified as NPs [8]. NPs have attracted intensive attention of scientific community and exist in the form of solid particles or particulate dispersions which have dimensions in the range of 1–100 nm [9,10]. As a result, the physical, chemical, optical, and magnetic properties of NPs vary drastically as compared to bulk material promising their wide range applications [11].

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With size reduction, the nanoscale shape and size impact the properties of NPs. For example, for metal NPs, the mean free path of an electron can be longer than the material dimensions. The distance can be so small that the energy required to charge a nanostructure can be similar to the thermal energy. Fig. 1 illustrate the conversion of bulk material into NPs followed by an increase in the important associated properties.

NPs can be classified as 1D, 2D and 3D depending on the dimensions within which a free electron is enabled to travel [12]. NPs are either metallic or non-metallic in nature. Non-metallic NPs are either organic or inorganic substances. Metallic NPs are extensively used due to their properties of electrical conductivity and density. Other important properties include high surface-area to volume ratio, uniformity, low sintering temperatures due to shape and size dependant melting point depression, magnetic properties, hardness, potential for superior functionalization and the presence of collective electron oscillations enabling the surface plasmon resonance phenomenon [11,13–15].

2.1. Applications

The industry based on nanomaterials is emerging; NPs are finding applications in almost every field. For example in medicine and biological applications [16], NPs are used in tissue engineering [17,18], multicolour coding in biological assays [19], cancer therapy [20–22], detection of proteins [23], and biomedical imaging for early detection of disease [24], respectively. In ophthalmology, NPs are effectively used because they allow the targeted drug delivery to reach the specific locations in the eye [25]. The industrial applications of NPs based printed and wearable electronics are growing rapidly [26,27]. NPs and NPs based sensors are used to detect the environmental pollutant such as toxins, heavy metals in air [28,29], for water purification [30–32] and in food safety to detect the food contaminants [33,34]. NPs also serve as catalytically active materials in modern chemical industry [35–37]. NPs are also used in smart, wearable sensors for human health monitoring [38–40]. For example, a highly sensitive pressure sensor utilizing silver (Ag) NPs electrode arrays has been demonstrated for detecting the human body movements [41]. Other applications involve temperature monitoring, chemical sensors for health monitoring, flexible energy storage devices, UV protective textiles, strain, and pressure sensors, respectively [32,42–45]. Moreover, the incorporation of metallic NPs in solar cell applications not only improves the power conversion efficiency but it also adds value in the cost-effectiveness of the process for large scale production [46,47]. The trapping of light at the interface increases due to surface plasmon resonance; a special characteristic of Ag and gold (Au) NPs that leads to the generation of hot-electrons responsible for the photo current in solar cells [48–50]. NPs based thin films also demonstrate potential applications in microelectronics,

microelectromechanical systems (MEMS), lab on chip (LOC) systems, microfluidic devices, optical component coatings, solar cells, optical & medical devices, organic field effect transistors (OFETs) and in transparent conductor [51–54]. Some application areas of NPs and NPs based thin films are illustrated in Fig. 2.

2.2. Deposition of NPs and thin films

The deposition of NP inks on a surface can be carried out in many ways. Different droplet-based techniques have been used to transfer NPs on desired substrate including inkjet, spray, aerosol jet and electro-hydrodynamic jet-based methods [59–61]. These techniques will be discussed briefly in the sections below.

2.2.1. Inkjet printing

Inkjet printing is recognized as an efficient method for direct deposition of functional inks on flexible, heat sensitive substrates [62]. Three steps are integral to inkjet printing; (1) pre-printing step to optimise the ink and substrate properties, (2) the actual printing step, and (3) the post printing sintering step to optimise the metal's conductivity [63]. Inkjet printing is a non-contact technology that operates for two ejection modes namely continuous inkjet printing and drop-on-demand (DOD) inkjet printing. The DOD printers are either piezoelectric or thermal inkjet printers [59,62].

Piezoelectric inkjet: Piezoelectric inkjet printing pushes droplets out through a nozzle using a mechanical force. A piezoelectric crystal in the chamber undergoes a distortion when an electrical signal is applied, reducing the chamber volume, thereby forcing the ink through the print head nozzle [64]. The attraction to piezoelectric approach is that the deposition is purely mechanical and avoids any of the complications associated with high temperature processes. The droplets have a diameter of 10–30 μm or have a volume of 1–20 pL. The velocity of the ejected droplets is 1–20 ms^{-1} . The stand-off distance between the printer and the nozzle is typically 1 mm. In many systems, multiple nozzles aligned in a linear configuration are used to execute a reliable printing step. Also, the stability of the NPs ink distribution needs careful attention and tailoring to a specific substrate [65,66]. The disadvantage rests with the viscosity limitations up to 20 cP.

Thermal inkjet: Thermal inkjet pushes droplets out through a nozzle by forming a thermally induced bubble in the ink on microsecond time scale. The triggering signal passes through the heating element that forms vapour bubbles. These vapour bubbles cause an expansion of the liquid in the nozzle resulting in the droplet ejection [59]. A thermal bubble actuator generates heat which can cause a damage to the print-head at higher temperatures for a prolonged printing duration [62]. The

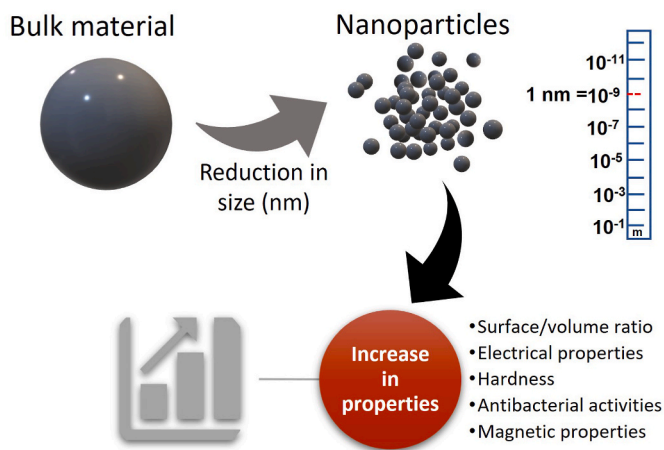


Fig. 1. Schematic illustration of reducing the size of bulk material into nano sized particles and corresponding increase in the properties.

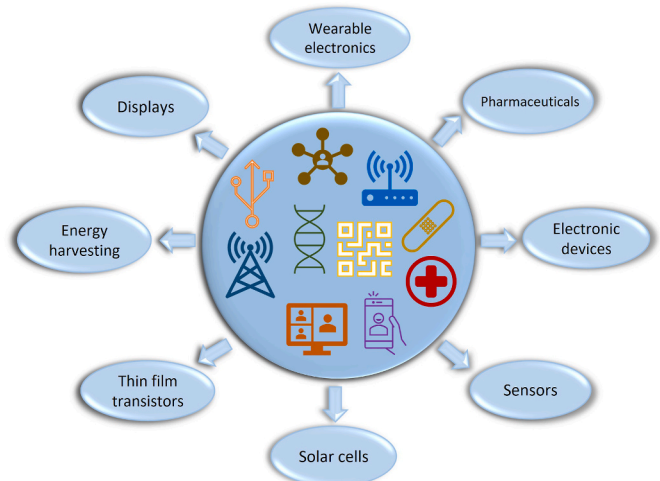


Fig. 2. Applications of nanomaterials in different areas [55–58].

advantages of thermal inkjet are its cost, efficiency, and limited resolution to micrometres. However the disadvantage rests in the thermal impact of the heater which can be undesirable for many inks and the printheads [67,68].

2.2.2. Aerosol jet printing

Aerosol is defined as a suspension of liquid or solid particulates in a gas carrier. Aerosol jet printing produces an aerosol form of the material with an approximate particle diameter of 1 μm . Aerosol jet printing starts with a nebulization step that comprises of the generation of aerosols. There are two main methods for aerosols generation depending on the ink viscosity: (a) ultrasonic and (b) pneumatic nebulization [69, 70]. In former method, an ultrasonic atomizer used to produce high frequency pressure waves generated by a piezoelectric transducer to create ink droplets with a viscosity ranging from 1 to 5 cP [59]. In later method, the high velocity carrier gas passes through a pneumatic atomizer is capable of generating ink droplets through a Venturi effect for a larger range of viscosity from 0.7 to 2500 cP [69,71].

2.2.3. Electro-hydrodynamic printing

Electro-hydrodynamic jet printing (E-jet printing) is a droplet based direct-writing technique. Electro-hydrodynamic printing employs electric fields, rather than acoustic or thermal energy, to create fluid flow through microcapillary nozzles to deliver the ink on a substrate [59]. When voltage is applied between the nozzle and the conducting substrate, the electric field causes the charge accumulation on the pendant droplet coming out of the nozzle. This charge accumulation results in a shear stress deforming the ink meniscus to a conical shape known as Taylor-cone [72,73]. When the electric potential of the meniscus exceeds above a certain value, the surface tension is overcome by electrostatic forces causes an ejection of thin jet from the apex of the Taylor-cone onto a substrate [73,74]. The size of the ejected droplets can be up to 10 or 1000 times smaller than the diameter of the nozzle.

2.2.4. NPs based thin film deposition techniques

NPs based thin film deposition techniques are classified into two categories, (1) physical methods and (2) chemical methods for thin film preparation. Physical techniques are subdivided into vacuum evaporation and glow discharge processes. Chemical deposition methods are subdivided into gas and liquid phase categories [75–79]. A detailed classification of physical and chemical deposition techniques of thin film is presented in Figs. 3 and 4 respectively. It should be noted that the properties of NPs thin films i.e., particle size, film thickness, surface roughness, film morphology and the mechanical properties are strongly affected by the type of deposition method used. Additionally, the deposition parameters e.g., deposition rate, current, temperature and atmosphere also contribute to the physical, chemical, mechanical, and structural properties of thin NPs films, respectively.

3. Post processing

After deposition of NP inks, a post processing step is essential. Generally, the functional NP inks consist of metal NP dispersions in the solvents. To avoid the aggregation of these NPs in the dispersion, metal-particle bindings to thiols and polymers (capping agents) are used as stabilizing agents [80,81]. After printing, the metal NPs do not exhibit the desired electrical properties due to the presence of stabilizing agents and solvents. The bond between the particles and the ligands must be broken to remove the polymeric ligands since the presence of an organic layer (even few nanometres thin) can hinder the electronic mobility between the metal particles [80]. The removal of polymeric ligand results in the promotion of metal-to-metal atomic diffusion between the particles thereby enabling the restoration of the electrical conductivity of the printed pattern. Therefore, sintering step is important to create the metal-to-metal contact for desired improved electrical properties.

3.1. Sintering techniques

Different sintering methods are used including thermal [82–84],

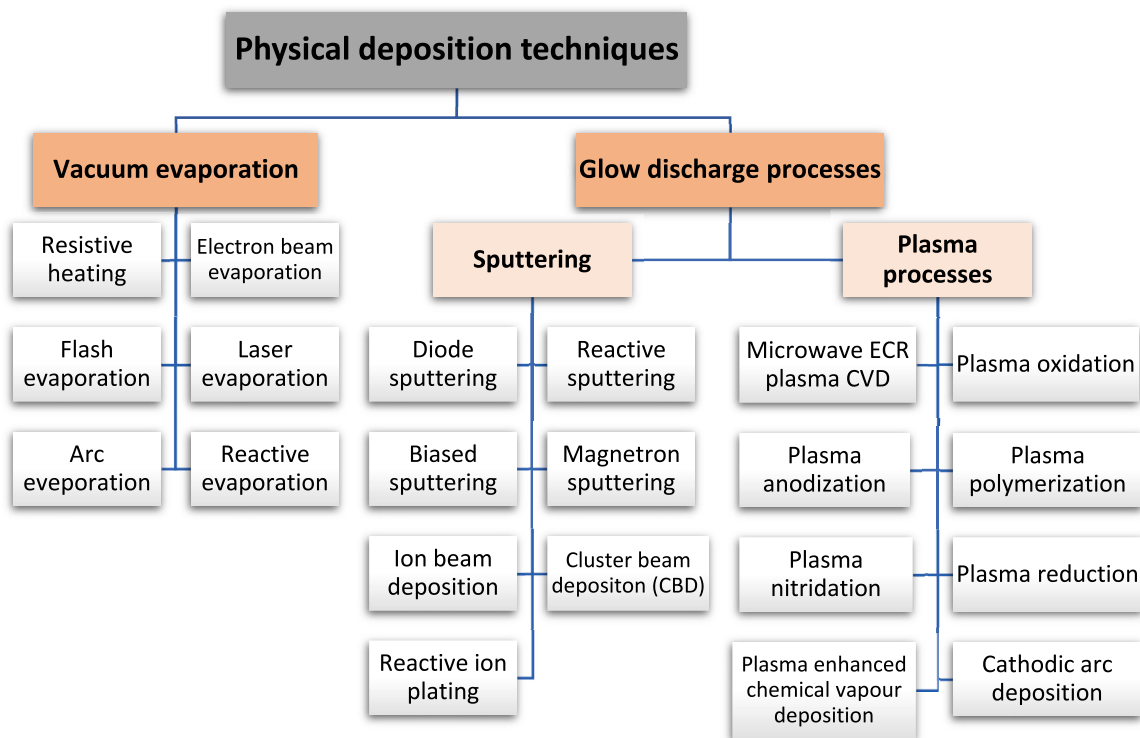


Fig. 3. Classification of NPs thin film deposition based on physical deposition techniques [75–79].

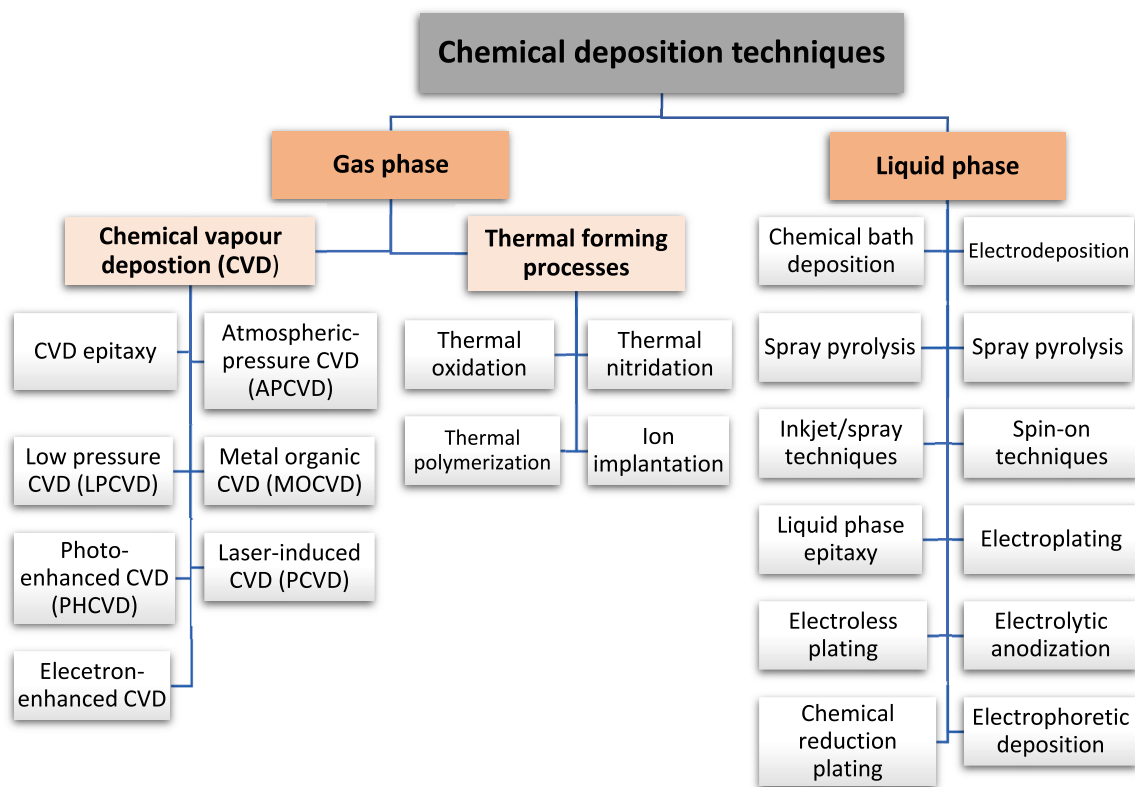


Fig. 4. Classification of NPs thin film deposition based on chemical deposition techniques [75–79].

intense pulsed light (IPL) [82,85,86], plasma [61,87,88], electrical [89,90], and laser sintering process [91–97]. The removal of the solvent using thermal sintering is the simplest approach, but it is not compatible with polymer substrates due to their high heat sensitivity. The heat sensitive printing substrates could suffer from shrinkage when exposed to high sintering temperatures. IPL or photonic flash sintering is performed with electromagnetic radiation which lies between the UV and IR radiation on the wavelength spectrum [2]. However, the pulse duration of several millisecond may thermally damage and deteriorate the properties of the substrate. In electrical sintering, the NPs film is sintered by applying a voltage over the printed pattern that allows the flow of electric current through the circuit and the local heating is possible [98].

3.2. Laser post processing

Lasers are actively used for sintering and fabrication of conductive patterns using different NPs based inks including metal, ceramic, polymer, semiconductor and oxide-based NPs [10]. Lasers are preferred as they offer several advantages including high peak irradiance, high precision, facile programming, compatibility with heat sensitive substrates, ambient-based manufacturing, reproducibility, and reliable hardware. The localized heating effects are desirable for selective patterning and sintering of conductive materials for printed electronics where the laser pulse duration and the wavelength are the key factors in determining the laser induced effects. The long exposure time and high powers of CW laser required for sintering results in the complete melting of NPs which on re-crystallization after cooling causes a low electrical resistivity. However, heating, followed by a cooling process with a CW laser is often accompanied with cracking, delamination, thermal damage, and low resolution of the printed patterns.

The issues with a CW laser can be overcome with pulsed laser processing where the application of short and ultrashort laser pulses have

gained much attention over the last two decades. Ultrashort laser processing provides a precise and reconfigurable solution for the issues that arise in CW and longer pulse lasers in a highly effective manner without impacting the nearby component and surfaces in a device. The pulse duration of an ultrashort laser is much shorter, typically in the sub picosecond domain [99]. Irradiating the material with ultrashort lasers could enable a non-thermal structural transition since the pulse durations is shorter as compared to the thermal coupling time between the electrons and phonons [100]. Fig. 6 illustrate the concept of variations in laser pulse width on different temporal scales. A salient feature of ultrashort laser material interaction is that only the electrons are excited first by photon-electron interaction within the pulse duration, leaving the lattice temperature unchanged. At this stage, the electrons establish the highest temperature and lattice is considered cold. These two temperature phenomena are well described by two temperature model (TTM) [101,102] considering the interaction of ultrashort laser with materials.

3.3. Sintering theory

The driving force for sintering is the reduction in the total interfacial energy. According to sintering theory of powders, it is classified in three categories (1) solid state sintering, (2) liquid phase sintering and (3) vapour assisted sintering [103]. Sintering normally occurs in three sequential overlapped stages referred as the initial, intermediate, and final stages [104,105]. The model for the initial stage consists of two equally sized spheres which are in contact. The classical sphere-to-sphere model states that the sintering process begins with a rapid neck formation followed by a neck growth process [106]. A rapid formation of inter-particle necking continues until the radius of the neck reaches to a value of $\sim 40\text{--}50\%$ of the particle radius [104,107]. The neck growth may occur due to different type of material transport including viscous flow, surface diffusion, grain boundary diffusion and

volume diffusion [104,108]. The neck formation between the particles of radius a is supposed to be circular in cross section with a neck radius x and can be written as [109];

$$\left(\frac{x}{a}\right)^n = Bt \quad (1)$$

Where, x/a is the ratio of neck radius to the particle radius, n is a mechanism characteristic exponent that is depended on the mass transport process, B is a constant depending on particle size, temperature, geometric and material terms and t is the sintering time, respectively. The exponent n is defined by the sintering mechanism. Therefore, one can study the driving mechanism for neck growth by estimating the value of n [108]. The intermediate stage implies that the inter-particle necks are larger than one-third of the particle size but less than half the particle size. The intermediate stage corresponds to pore rounding and the grain growth where a considerable densification up to $\sim 93\%$ of the relative density occurs before the isolation of the pores. As the sintering proceeds, the pore channels disconnect and isolated pores are formed with a grain growth [104]. At final stage, the densification of isolated pores occurs, and the pores collapsed into closed spheres. At least 7–8% of porosity is eliminated where the pores are no longer open to external surface [1]. Coble proposed two microstructure models for the shape change of pores: the channel pore model for intermediate and the isolated pore model for the final stage, respectively [104,110]. During all these stages, the atoms move by several transport mechanisms including surface diffusion and grain boundary diffusion to change the microstructures as illustrated in Fig. 5. The diffusion of atoms results in either densification or the coarsening of the microstructures.

4. Ultrashort laser processing of NPs

4.1. Laser nano-welding (joining) of NPs

Ultrashort laser sintering is a promising tool to selectively sinter almost all types of NPs with minimal heat affected zone and thermal damaging. First, we will start with the nano-joining of only two metal NPs as shown in Fig. 7a. Kim et al. reported laser induced nano-welding of 13 nm Au NPs on carbon coated copper (Cu) TEM grids using 532 nm ps laser pulses of 30 ps pulse duration at 10 Hz [111]. It was observed that the laser irradiation initially caused the Au NPs to associate with each other and further irradiation causes the neck formation resulting in the welding. High resolution TEM images revealed welding of Au NPs, confirming well-connected single phase FCC crystalline structure of Au because of localized heating after laser energy absorption.

As stated previously, functional NP inks are generally coated with binders and stabilizers to avoid agglomeration. Huang et al. investigated

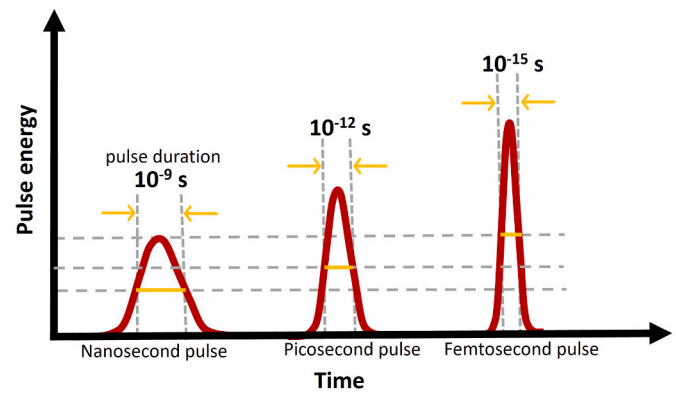


Fig. 6. A conceptual illustration to differentiate between the pulse duration of short and ultrashort lasers.

the joining behaviour of polyvinylpyrrolidone (PVP) coated Ag NPs when exposed to femtosecond laser pulses using 800 nm wavelength, 35 fs pulse duration at 1000 Hz repetition rate [112]. The formation of an intermediate hydrogenated amorphous carbon (α -C:H) coating on individual Ag NPs takes place at low laser fluence due to decomposition of PVP. This effect is associated with the ejection of electrons at hot spots created by the trapping of plasmons. When the fluence is increased between 200 and 300 μJcm^{-2} , the softening of Ag lattice followed by electron emission occurs. This leads to the removal of α -C:H and results in the local welding of nearby Ag NPs. The carbonaceous shell is removed, and a metal neck formation starts. Fig. 7b exhibits the TEM (left) and HRTEM (right) images of Ag NPs sintered at (a & b) 90 μJcm^{-2} and (c & d) 200 μJcm^{-2} for 20 min. Other studies for nano-welding/joining of NPs are reported in Refs. [5,100,113–115].

4.2. Ultrashort laser sintering

The fabrication of submicron-sized features is inhibited by large heat diffusion and light diffraction limit when using CW lasers. For this purpose, short or ultrashort pulsed lasers are commonly employed since they offer small heat affected zone and reduced damage to the substrate compared to CW lasers. The thermal behaviour is attributed to the thermal diffusivity of the substrate material defined as the rate of heat propagation and heat absorption in the material. Thermal diffusivity becomes an important factor to consider the thermal response of the substrate when using CW and short pulse lasers processing. Roy et al. explored the sintering mechanism of Cu NPs using femtosecond, nanosecond and CW lasers for a range of parameters including film thickness,

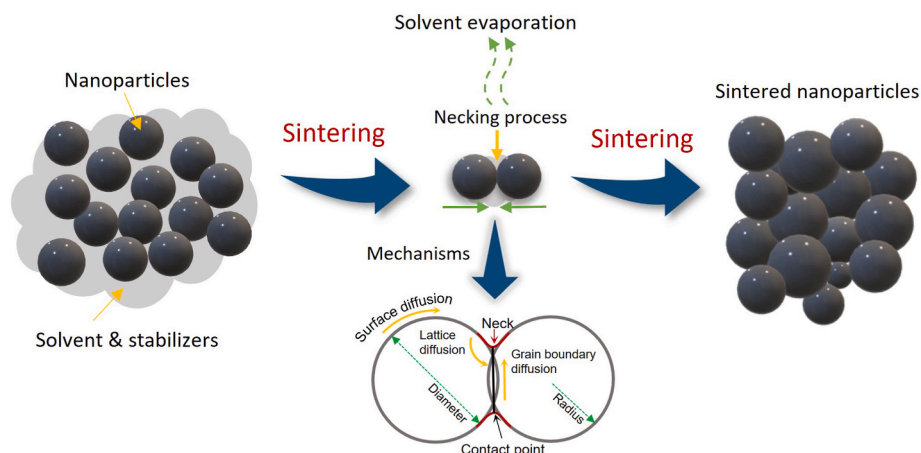


Fig. 5. Flow diagram of a typical sintering process of NPs.

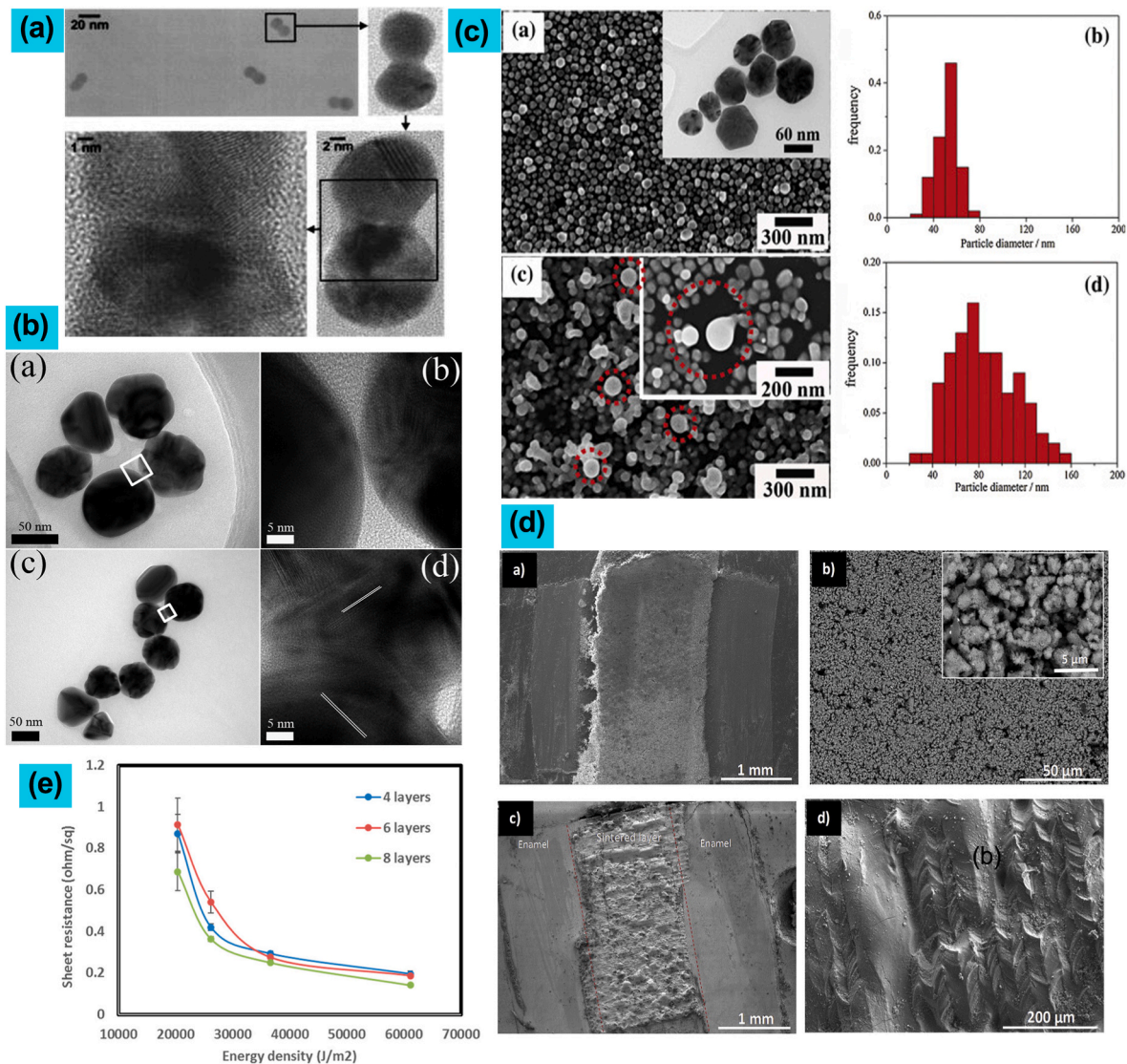


Fig. 7. (a) The HRTEM images of Au NPs nano-welding after exposing with picosecond laser pulses at 0.2 mJ for 10 min (Reprinted from ref. [111], Copyright © 2005, American Institute of Physics), where (b) shows the joining of Ag NP using femtosecond laser (Reprinted from ref. [112], Copyright © 2012, AIP Publishing). Figure (c) presents the SEM and TEM (inset) images of Ag NPs on Si wafer before and after femtosecond laser sintering at a fluence of 7.2 mJcm^{-2} for 10 s (Reprinted from ref. [117], Copyright © 2015, Elsevier), (d) provides the SEM images of fluorapatite (FAP) powder containing Fe₂O₃ NPs on an eroded bovine enamel before (a & b) and after (c & d) femtosecond laser sintering (Reprinted from ref. [118], Copyright © 2018, Elsevier). (e) Presents the variations in the sheet resistance as a function of energy density of femtosecond laser sintered Ag NPs for strain sensor application (Reprinted from ref. [119], Copyright © 2018, American Chemical Society).

thermal properties, exposure time and laser power [93]. The processing window was categorized into no sintering effects, weak sintering, good sintering, and ablation/evaporation, respectively. The effect of substrate thermal diffusivity on sintering threshold was only dominant in CW where the high laser power is required as compared to nanosecond and femtosecond lasers. It was found that the femtosecond laser (800 nm, 5 kHz, 100 fs) needed the minimum threshold fluence as compared to other lasers. A comparison study of sintering Ag NPs on polyethylene naphthalate (PEN) using CW, pulsed nanosecond and picosecond laser operating at with 532 nm and 1064 nm wavelength is described in Ref. [116]. The simulation predicts a reduced thermal diffusion on PEN for picosecond laser pulse where in case of nanosecond laser, the heat affected zone was restricted to few microns below the surface. The damage and the heat affected zone found dependant on the incident laser power of the CW laser. The sintering efficiency was higher with CW and nanosecond laser, but an impact of damage was directly related to the incident laser power. The onset of damage was negligible with

picosecond laser, but efficiency of sintering process was challenging. The absorption of Ag NPs at IR and green wavelength is responsible for the electrical behaviour of sintered lines. The strong absorption of Ag at 532 nm wavelength causes a solid superficial layer preventing the heat reaching at the bottom layer of the material. Therefore, the NPs near the interface are not completely in a molten state. This effect was pronounced with picosecond laser where sintering was not effective at 532 nm wavelength.

As discussed previously, ultrashort pulse laser enables the fabrication of submicron sized features due to the advantage of reduced heat accumulation effects in laser processed area. The selective melting of Ag NPs on polydimethylsiloxane (PDMS) substrate was carried out to fabricate high resolution, submicron features using 780 nm laser of 100 fs pulse duration [120]. Femtosecond laser sintering resulted in uniform metal lines of 380 nm width, smaller than the laser wavelength (780 nm). The formation of such features was due to the combined effect of ultrashort pulse duration and the low thermal conductivity of NPs that

limits the sintering area within the focal spot by reducing the thermal diffusion effects. Moreover, the presence of enhanced electric field around the NPs also facilitates the efficient absorption of laser energy [120].

The theoretical studies supported by experimental evidence are discussed in literature to understand the sintering mechanisms of NPs when irradiated with ultrashort laser pulses [93,121–123]. It was observed that liquid-phase sintering of NPs occurs when femtosecond laser pulses with higher repetition rate are employed for sintering of Cu NPs [122,123]. Whereas in nanosecond laser sintering, the sintering can be triggered either due to coalescence or by melting process depending on the laser fluence [93]. In a study, the application of overlapped femtosecond laser pulses at low fluence is inspected to observe the localized heating and melting of Ag NPs on silicon (Si) wafer [117]. The localized surface plasmons generated hotspots increase the effective heating and melting where the Ag NPs in specific symmetric plasmonic structures can be heated up to their melting point at around $\sim 7.2 \text{ mJcm}^{-2}$ (Fig. 7c). It was found that the generation of spherical particles because of laser sintering is independent of the exposure time, but the interaction is mainly affected by the laser fluence and the presence of localized structures on the NP film [117].

Cheng et al. [123] investigated the ultrashort laser induced melting of Cu NPs experimentally and theoretically initiated by 800 nm, 100 fs laser pulses. A one-dimensional TTM model was used to solve the heat transfer equations including the temperature dependant material

properties, including the extended Drude model for dynamic optical properties, and the thermophysical properties, respectively. Another finding reports a simulation study consisting of TTM model with dynamic optical properties (Drude model) and a rapid phase change for melting of Ag NPs for ultrafast laser induced melting [122]. The simulation results were compared with existing experimental data in the literature and later evaluated and concluded with the experimental findings of femtosecond laser (1064 nm wavelength, 300 fs pulse width, 1 MHz repetition rate) sintered Ag NPs of 5 nm size. A complete sintering of Ag NPs into bigger grains (150–250 nm) was obtained at the higher fluence of 7.7 mJcm^{-2} , where an incomplete sintering is obtained at decreased fluence such as 3.5 mJcm^{-2} . This experimental value of melting fluence such as 7.7 mJcm^{-2} agreed with the estimated multi shot fluence (8.6 mJcm^{-2}) from simulation. The resistance of sintered Ag lines reduced from $\sim 715 \Omega$ to $\sim 344 \Omega$ measured with 2-point probe [122]. Recently, the fabrication of conductive Ag lines through a combination of laser induced forward transfer (LIFT) printing and laser sintering using same 532 nm ps laser of 10 ps pulse duration is demonstrated in Ref. [95]. The lowest resistivity of $1.4 \times 10^{-7} \Omega\text{m}$ was achieved at laser energy of $1.3 \mu\text{J}$, scanned at 7 mms^{-1} and 0.8 MHz, respectively. The numerical study was performed using a classical axisymmetric TTM model to evaluate the electronic and lattice distribution during laser sintering process. TTM results showed a heat accumulation phenomenon which increases with an increase in number of pulses and pulse energy but the overall thermal damage on the substrate

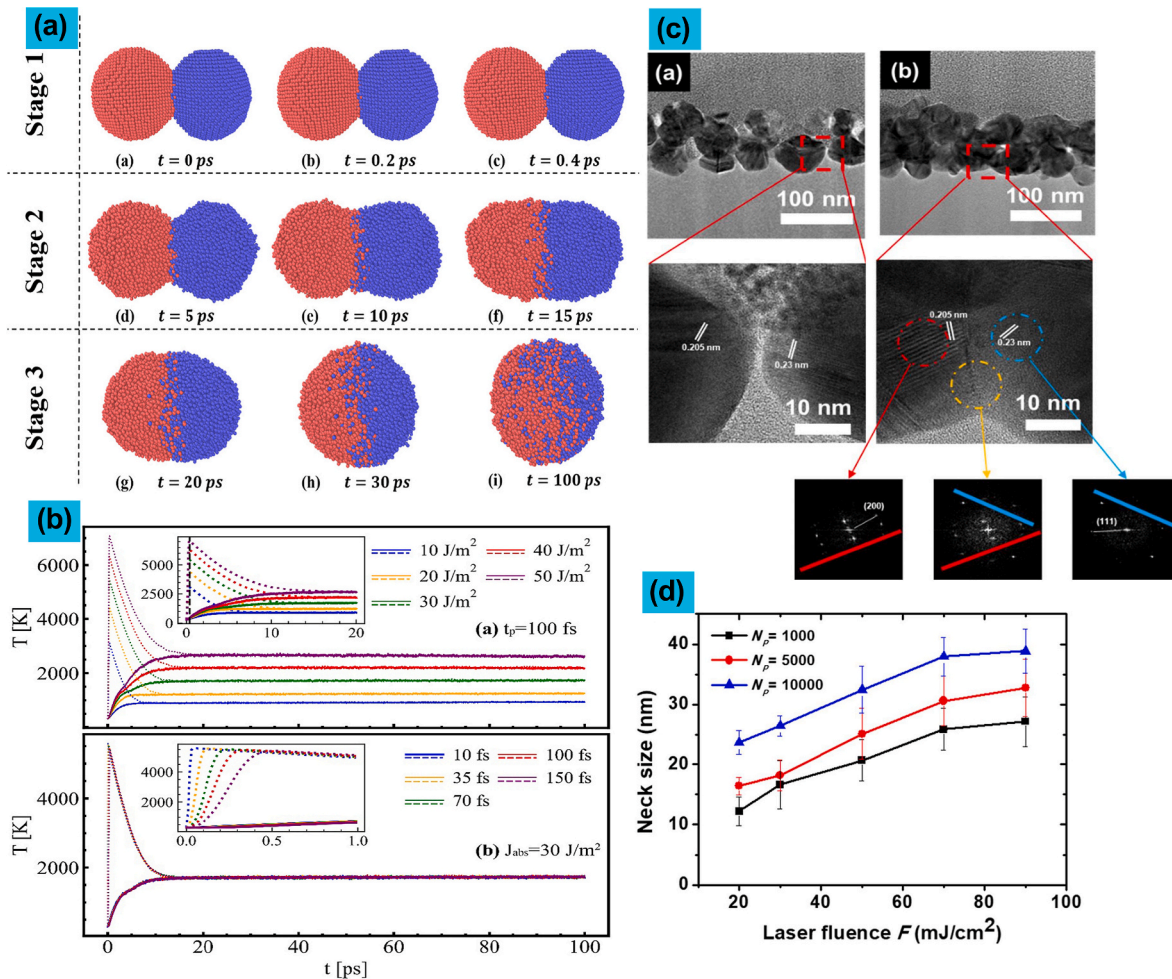


Fig. 8. (a) Visualization of femtosecond laser sintering process of two Al NPs from atomistic snapshots where (b) shows the temporal evolution of electron and lattice temperature (T_e & T_l) at different laser fluence and different pulse duration (Reprinted from ref. [125], Copyright © 2022, Elsevier). (c) high magnification TEM and selective area electron diffraction (SAED) images of Ag NPs before and after femtosecond laser sintering where (d) describes the relationship between the neck size and laser fluence of sintered Ag patterns (Reprinted from ref. [121], Copyright © 2020, Elsevier).

was insignificant. Ekici et al. simulated the heating process of Ag NPs in an aqueous medium with femtosecond laser pulses where the accuracy of his axisymmetric TTM model found within the range of 5–15% [124].

A recent study combines the molecular dynamics simulation at atomistic scale and TTM model at continuum scale to understand the sintering process of aluminium (Al) NPs [125]. The diameter of NPs is 6 nm and laser pulse width are 100 fs. The Al NPs coalesce to minimize the surface energy and the time scale of this sintering process is given in Fig. 8a. The calculated time at which the two Al NPs coalesce into a spherical one is about 30 ps. The temporal evolution of electron and lattice temperatures, at (a) same pulse width but at different laser fluence and (b) constant laser fluence but varied pulse duration, is provided in Fig. 8b. In a study reported by Noh et al. for the comparison of nanosecond and femtosecond laser sintering of Ag NPs on PET substrate, the numerical simulation using finite element method was carried out to estimate the electron and lattice temperature at different process conditions [121]. The calculations were carried out for a layer of Ag NPs with the properties close to bulk values which is a reason of the underestimated temperature in TTM. However, the TTM results were interpreted by considering the data from differential scanning calorimetry (DSC) results. No heat accumulation effects were observed during pulse overlap where the heat relaxation time was large (several μ s). The maximum temperature was below the melting temperature of Ag NPs, consistent with the SEM observation, revealing that necking happened mainly by solid-state diffusion process (Fig. 8c).

4.3. Laser sintering for biomedical applications

The ultrashort nature of a femtosecond laser is responsible for overcoming the thermal effects making it a strong candidate for hard tissue engineering [126]. A recent approach of exogenous mineralization of hard tissues such as dental enamel using photo absorptive minerals and femtosecond laser pulses is proposed in the literature [118]. A layer of Fe^{3+} -ions doped calcium phosphate mineral such as fluorapatite (FAP) powder containing Fe_2O_3 NPs is applied on an eroded bovine enamel followed by a femtosecond laser sintering process. The IR (1040 nm) laser with a pulse width of 150 fs at 1 GHz repetition rate is used to sinter the FAP paste containing Fe_2O_3 NPs where the incorporation of Fe^{3+} ions in FAP increases the linear absorption of IR laser. A 20 μ m thick sintered layer is formed due to densification of surrounding calcium phosphate crystals without affecting the underlying surface. Fig. 7d shows the SEM image of FAP layer before and after laser sintering on tooth enamel. The resurfacing of teeth with calcium phosphate minerals was also demonstrated using a femtosecond laser [127]. The dental tubules coated with Er^{3+} -ions doped and undoped calcium phosphate were sintered with a 1520 nm, 100 fs laser at 2.5 GHz where the addition of Er^{3+} -ions increases the laser energy absorption at this wavelength. A similar study of femtosecond laser sintering (190 fs, 1045 nm) of calcium phosphate biomaterials doped with Fe^{3+} -ions is reported for surface restoration of hard tissues such as bone or enamel in Ref. [128].

Femtosecond laser sintering also enables the fabrication of integrated microfluidic Surface Enhanced Raman Scattering (SERS) substrates using Ag NPs as reported in Ref. [129]. Ag NPs were injected into fused silica based microfluidic channel. Femtosecond laser of 1030 nm wavelength, 370 fs pulse duration operating at 1 MHz was focussed through the top fused silica layer to sinter the embedded Ag NP paste in the microfluidic channel. It was found that the formation of a rough surface due to particle necking led to the formation of hot spots important for SERS phenomenon [129]. Moreover, ultrashort laser sintering is potentially used to sinter NPs specifically on heat sensitive materials. This approach is used to prepare the Ag NPs based bandage for healthcare applications [119]. Aerosol jet printed Ag NPs on a commercially available bandage made of thin polyurethane membrane coated with acrylic adhesive was used for the fabrication of a strain sensor. A femtosecond laser of 780 nm wavelength and 100 fs pulse

width was employed to sinter the Ag NPs for the fabrication of a strain sensor. Fig. 7e shows the variations in sheet resistance as a function of laser energy density for different layers of Ag NPs that is observed to be decreasing with increasing energy density.

4.4. Laser reductive sintering of metal oxides

Femtosecond lasers are also reported for laser reductive sintering of metal oxides for example, cobalt oxide (CoO) [130] and copper oxide (CuO) [131]. Selective laser-induced reduction of air stable CuO NPs on flexible polyimide substrate is demonstrated using femtosecond laser reductive sintering process operating at high repetition rate of 76 MHz [132]. The laser pulse duration was 100 fs with a wavelength of 1030 nm. It was proposed that the laser exposure on CuO NPs may trigger the thermochemical reduction reaction converting CuO into Cu NPs. Additionally, the thermal effects generated by high repetition rates caused the heating of CuO NPs up to 260 °C. As a result, the surface atoms start moving and a neck formation occurs under the action of van der Waals forces [132].

4.5. Ultrashort laser sintering of metal oxides

The laser sintering mechanism is strongly affected by the optical properties of NPs as the laser wavelength must be chosen in a way where the maximum laser energy is absorbed by the NPs. It is easier to find the maximum absorption for metal NPs since their plasmon resonance peaks are identified in the visible region of spectrum [133]. However, sintering of transparent conducting oxides (TCO) NPs such as indium tin oxide (ITO) NPs is not straight forward due to their transparency in the visible spectral region. The optical measurements of ITO NPs on PET substrate were performed to select the most appropriate region of spectrum where maximum absorption is achievable [134]. ITO NPs showed strong absorption in near IR region where the PET was transparent. The gravure printed ITO NPs ink was subjected to ultrashort laser sintering using 6 ps laser pulses. Fig. 9a shows a significant increase in the grain size in the range of 100–200 nm and its dependence on laser fluence is given in Fig. 9b.

In a separate study, highly transparent thin films of ZnO NPs were prepared on quartz using spin coating process and exposed to a UV laser of 12 ps pulse duration at 80 MHz [135]. The laser wavelength was selected in the 355 nm as the ZnO NPs showed maximum absorption in UV region due to band gap which is nearly 3.3 eV. The as deposited ZnO films were not conductive, but the selective ultrashort laser annealing and direct writing of conductive patterns enabled the fine tuning of electrical and physical properties of ZnO. Picosecond laser pulses also used for annealing of multilayer structures consisting of germanium (Ge) and Si multi nanolayers. Ge layers were partially crystallized where the α -Si was unaffected without obvious intermixing of Ge and Si [136]. Additionally, a recent study reports a simultaneous enhancement in the electrical properties and optical transparency of 50 nm and 100 nm thick ITO films on PET substrate as a result of green femtosecond laser scanning [137]. Fig. 9c demonstrates the AFM topographic images of 50 nm (a & b) and 100 nm (c & d) ITO films on heat sensitive PET substrate. The average grain size of ITO films enhances after laser annealing. Fig. 9d exhibits the TTM simulation results performed considering the PET interface which revealed that the estimated lattice temperature and temperature at the interface was below the melting temperature of ITO films.

4.6. Ultrashort laser crystallization of thin films

Other studies of ultrashort laser annealing of NP thin films without melting the underlying substrate are recently reported in the literature [7,138–140]. It was observed that the degree of solid-state induced crystallization as a result of ultrashort laser annealing in the Au thin films is strongly dependent upon laser wavelength as observed for IR,

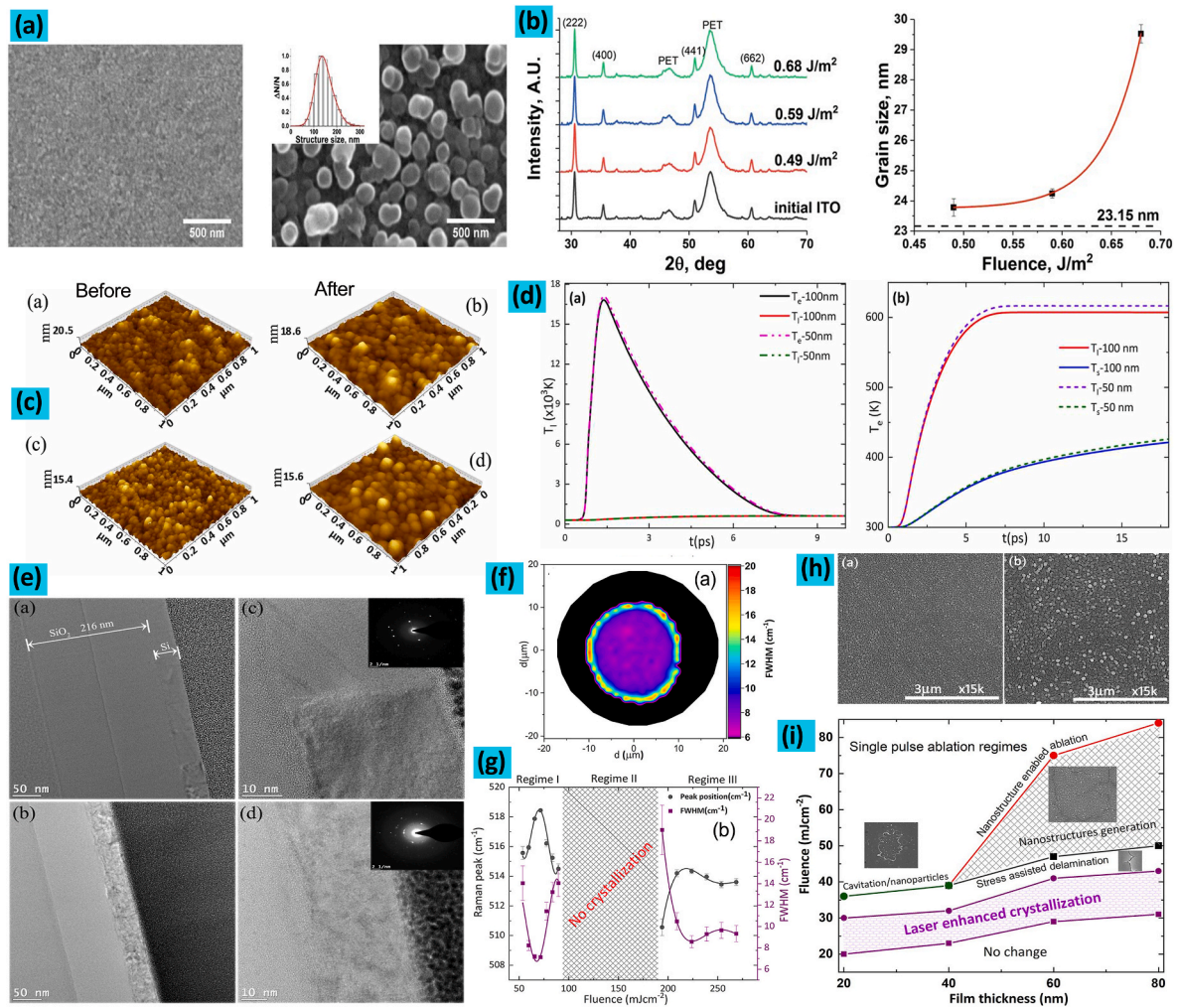


Fig. 9. Ultrashort laser annealing using femtosecond laser pulses in thin NPs films. (a) SEM images describing the surface morphology of gravure printed ITO ink before and after laser sintering, where (b) shows the XRD diffractogram and grain size with increasing laser fluence, respectively (Reprinted from ref. [134], Copyright © 2019, Nature Portfolio). (c) Shows the AFM topographic images of 50 nm (a & b) and 100 nm (c & d) ITO films before and after femtosecond laser scanning where (d) provides theoretical validation of low fluence crystallization below the melting threshold of ITO films using TTM modelling (Reprinted from ref. [137], Copyright © 2021, IOP Publishing Ltd). (e) Presents the TEM images of amorphous Si before (a & c) and after (b & d) IR laser crystallization where (f) shows the Raman mapping of crystallized region at 72 mJcm^{-2} and (g) explains the Raman peak and FWHM variations versus laser fluence (Reprinted from ref. [138], Copyright © 2021, American Chemical Society). In (h) selective annealing of 40 nm Mo film before and after femtosecond laser irradiation is provided (Reprinted from ref. [139], Copyright © 2021, IOP Publishing Ltd) and (i) depicts the laser enhanced crystallization in 20 nm, 40 nm, 60 nm and 80 nm thick Mo films, respectively (Reprinted from ref. [7], Copyright © 2022, Elsevier).

green and UV laser pulses [140]. Moreover, the use of an additional layer such as molybdenum (Mo) could trigger the crystallization process of an amorphous Si (α -Si) as compared to the one where the metal layer is not used when subjected to femtosecond IR laser annealing [138]. Three regimes were probed where laser triggers the crystallization in α -Si at low fluence (Regime I), no crystallization (Regime II) at intermediate fluence, and crystallization at higher fluence (Regime III) as a result of laser annealing (Fig. 9g). Fig. 9e describes the cross-sectional TEM images of samples after crystallization at 72 mJ cm^{-2} (Regime I at low fluence) and at 269 mJ cm^{-2} (Regime III at high fluence), respectively. The direct, selective low fluence annealing of Mo thin film is shown in Fig. 9h where it is challenging to anneal a material of higher melting temperature deposited on low melting temperature substrates [139]. Because the application of high laser power could lead to melting of heat sensitive substrate on which the Mo film is coated. The detailed investigation of ultrashort laser processing on Mo thin films revealed a sequence of fluence dependant regimes where one processes changes to another by simultaneous crystallization due to laser induced annealing within the lower limit of fluence range [7]. Increasing the laser fluence

unveils the non-thermal and thermally induced ablation mechanisms in the material which must be considered when annealing NP thin films (Fig. 9i). Ultrashort laser pulse annealing especially for crystallizing semiconductor thin films has been reported by many researchers in the literature [141–145].

Sintering is a process in a number of ways where the absorption of laser energy facilitates the fusion of NPs. The resultant NP structures are improved with enhanced structural, mechanical, and electrical properties. The sintering mechanism is greatly affected by the type of heating source, operating conditions, and environments. For lasers, when the pulse duration is reduced from milliseconds to femtoseconds, the sintering mechanism changes from complete melting of NPs to coalescence of NPs triggered by a solid-state diffusion process. Laser fluence and the repetition rate are important parameters that impact the laser energy absorption, electronic excitations, and electron-phonon thermalization in the materials. At low fluence, sintering by interparticle neck formation can be initiated by the solid-state diffusion in ultrashort lasers. Increasing the fluence leads to liquid phase sintering but the overall thermal damage to the substrate is minimal. In summary, pulsed laser

sintering has a great potential to fabricate high resolution, highly conductive NPs thin film patterns on heat sensitive, flexible substrates. There is scope for new ways to sinter and crystallize materials in the future based on TTM concept where the higher electronic temperature can modify the material without increasing the temperature of the lattice.

5. Conclusions

This study highlights the progress and recent work reported in the field of laser sintering of NPs and NPs based thin films for flexible, printed electronics application. The conventional oven-based sintering and CW or longer pulsed laser sintering methods are not suitable for heat sensitive materials as they produce high heat affected zone. Ultrashort sintering has advantage in precise low temperature material processing with a low thermal budget. The high resolution, highly conductive patterns are achieved with ultrashort pulsed laser where sintering could be either triggered by a necking process due to solid state atomic diffusion at low power, or by the melting of NPs at higher powers and higher repetition rate resulting in the recrystallization of sintered film.

CRedit author statement

Ayesha Sharif: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft preparation, Writing - review & editing. **Nazar Farid:** Conceptualization, Data curation, Investigation, Methodology, Software, Validation, Resources, Writing - review & editing. **Gerard M. O'Connor:** Conceptualization, Validation, Resources, Project administration, Funding acquisition, Supervision, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] Z.Z. Fang, *Sintering of Advanced Materials*, Elsevier, 2010.
- [2] L. Mo, et al., Silver nanoparticles based ink with moderate sintering in flexible and printed electronics, *Int. J. Mol. Sci.* 20 (9) (2019) 2124.
- [3] G. Cicala, G. Bruno, P. Capezzuto, Chapter 9 - plasma deposition of microcrystalline silicon: the role of plasma-surface interaction on the microstructure, in: H.S. Nalwa (Ed.), *Handbook of Surfaces and Interfaces of Materials*, Academic Press, Burlington, 2001, pp. 509–543.
- [4] J. Poate, *Laser Annealing of Semiconductors*, Elsevier, 2012.
- [5] A. Hu, et al., Femtosecond laser welded nanostructures and plasmonic devices, *J. Laser Appl.* 24 (4) (2012), 042001.
- [6] K.P. O'Flynn, K.T. Stanton, Laser sintering and crystallization of a bioactive glass-ceramic, *J. Non-Cryst. Solids* 360 (2013) 49–56.
- [7] A. Sharif, et al., The role of fluence in determining the response of thin molybdenum films to ultrashort laser irradiation; from laser-induced crystallization to ablation via photomechanical ablation and nanostructure formation, *Appl. Surf. Sci.* (2022), 153315.
- [8] G. Schmid, *Nanoparticles: from Theory to Application*, John Wiley & Sons, 2011.
- [9] N. Strambeanu, et al., Nanoparticles: definition, classification and general physical properties, in: *Nanoparticles' Promises and Risks*, Springer, 2015, pp. 3–8.
- [10] I. Khan, K. Saeed, I. Khan, Nanoparticles: properties, applications and toxicities, *Arab. J. Chem.* 12 (7) (2019) 908–931.
- [11] D. Guo, G. Xie, J. Luo, Mechanical properties of nanoparticles: basics and applications, *J. Phys. D Appl. Phys.* 47 (1) (2013), 013001.
- [12] J.N. Tiwari, R.N. Tiwari, K.S. Kim, Zero-dimensional, one-dimensional, two-dimensional and three-dimensional nanostructured materials for advanced electrochemical energy devices, *Prog. Mater. Sci.* 57 (4) (2012) 724–803.
- [13] H.T. Phan, A.J. Haes, What does nanoparticle stability mean? *J. Phys. Chem. C* 123 (27) (2019) 16495–16507.
- [14] B. Akbari, M.P. Tavandashti, M. Zandrahimi, Particle size characterization of nanoparticles—a practical approach, *Iran. J. Mater. Sci. Eng.* 8 (2) (2011) 48–56.
- [15] W.J. Stark, et al., Industrial applications of nanoparticles, *Chem. Soc. Rev.* 44 (16) (2015) 5793–5805.
- [16] O.V. Salata, Applications of nanoparticles in biology and medicine, *J. Nanobiotechnol.* 2 (1) (2004) 1–6.
- [17] M. Fathi-Achachelouei, et al., Use of nanoparticles in tissue engineering and regenerative medicine, *Front. Bioeng. Biotechnol.* 7 (2019) 113.
- [18] A. Hasan, et al., Nanoparticles in tissue engineering: applications, challenges and prospects, *Int. J. Nanomed.* 13 (2018) 5637.
- [19] K.K. Jain, Nanodiagnostics: application of nanotechnology in molecular diagnostics, *Expert Rev. Mol. Diagn.* 3 (2) (2003) 153–161.
- [20] D. Zhi, et al., Targeting strategies for superparamagnetic iron oxide nanoparticles in cancer therapy, *Acta Biomater.* 102 (2020) 13–34.
- [21] A. Aghebbati-Maleki, et al., Nanoparticles and cancer therapy: perspectives for application of nanoparticles in the treatment of cancers, *J. Cell. Physiol.* 235 (3) (2020) 1962–1972.
- [22] A.A. Hassabo, et al., Anticancer effects of biosynthesized Cu₂O nanoparticles using marine yeast, *Biocatal. Agric. Biotechnol.* 39 (2022), 102261.
- [23] M.A. Santiago-Cordoba, et al., Nanoparticle-based protein detection by optical shift of a resonant microcavity, *Appl. Phys. Lett.* 99 (7) (2011), 073701.
- [24] X. Han, et al., Applications of nanoparticles in biomedical imaging, *Nanoscale* 11 (3) (2019) 799–819.
- [25] Y. Diebold, M. Calonge, Applications of nanoparticles in ophthalmology, *Prog. Retin. Eye Res.* 29 (6) (2010) 596–609.
- [26] J. Wiklund, et al., A review on printed electronics: fabrication methods, inks, substrates, applications and environmental impacts, *J. Manuf. Mater. Process.* 5 (3) (2021) 89.
- [27] S.P. Sreenilayam, et al., Advanced materials of printed wearables for physiological parameter monitoring, *Mater. Today* 32 (2020) 147–177.
- [28] S.P. Deshmukh, et al., Silver nanoparticles as an effective disinfectant: a review, *Mater. Sci. Eng. C* 97 (2019) 954–965.
- [29] L. Wang, et al., Nanoparticle-based environmental sensors, *Mater. Sci. Eng. R Rep.* 70 (3–6) (2010) 265–274.
- [30] M. Pinto, et al., Application of magnetic nanoparticles for water purification, *Environ. Adv.* 2 (2020), 100010.
- [31] T. Pradeep, Noble metal nanoparticles for water purification: a critical review, *Thin Solid Films* 517 (24) (2009) 6441–6478.
- [32] H.B. Ahmed, H.E. Emam, Seeded growth core-shell (Ag–Au–Pd) ternary nanostructure at room temperature for potential water treatment, *Polym. Test.* 89 (2020), 106720.
- [33] R. Jadhav, et al., An overview of antimicrobial nanoparticles for food preservation, *Mater. Today Proc.* (2022), <https://doi.org/10.1016/j.matpr.2022.07.045>. In press.
- [34] Y. Li, et al., Nanoparticle-based sensors for food contaminants, *TrAC, Trends Anal. Chem.* 113 (2019) 74–83.
- [35] D.T. Thompson, Using gold nanoparticles for catalysis, *Nano Today* 2 (4) (2007) 40–43.
- [36] D. Astruc, *Introduction: Nanoparticles in Catalysis*, ACS Publications, 2020, pp. 461–463.
- [37] C. Gao, F. Lyu, Y. Yin, Encapsulated metal nanoparticles for catalysis, *Chem. Rev.* 121 (2) (2020) 834–881.
- [38] A. Nag, S.C. Mukhopadhyay, J. Kosel, Wearable flexible sensors: a review, *IEEE Sensor. J.* 17 (13) (2017) 3949–3960.
- [39] H. Luo, B. Gao, Development of smart wearable sensors for life healthcare, *Eng. Regen.* 2 (2021) 163–170.
- [40] G. Appelboom, et al., Smart wearable body sensors for patient self-assessment and monitoring, *Arch. Publ. Health* 72 (1) (2014) 1–9.
- [41] Z. Zhou, et al., Supersensitive all-fabric pressure sensors using printed textile electrode arrays for human motion monitoring and human-machine interaction, *J. Mater. Chem. C* 6 (48) (2018) 13120–13127.
- [42] S. Mondal, Nanomaterials for UV protective textiles, *J. Ind. Textil.* 51 (4,suppl) (2022) 5592S–5621S.
- [43] K. Xu, Y. Lu, K. Takei, Multifunctional skin-inspired flexible sensor systems for wearable electronics, *Adv. Mater. Technol.* 4 (3) (2019), 1800628.
- [44] J. Zhao, et al., Recent advances in wearable self-powered energy systems based on flexible energy storage devices integrated with flexible solar cells, *J. Mater. Chem.* 9 (55) (2021) 18887–18905.
- [45] H.E. Emam, S. Zaghloul, H.B. Ahmed, Full ultraviolet shielding potency of highly durable cotton via self-implantation of palladium nanoclusters, *Cellulose* 29 (8) (2022) 4787–4804.
- [46] T. Temple, D. Bagnall, Optical properties of gold and aluminium nanoparticles for silicon solar cell applications, *J. Appl. Phys.* 109 (8) (2011), 084343.

- [47] M.A. Alkhalayfeh, A.A. Aziz, M.Z. Pakhruddin, An overview of enhanced polymer solar cells with embedded plasmonic nanoparticles, *Renew. Sustain. Energy Rev.* 141 (2021), 110726.
- [48] G.T. Mola, et al., Local surface plasmon resonance assisted energy harvesting in thin film organic solar cells, *J. Alloys Compd.* 856 (2021), 158172.
- [49] S. Xiao, et al., An ultra-low concentration of gold nanoparticles embedded in the NiO hole transport layer boosts the performance of p-i-n perovskite solar cells, *Solar RRL* 3 (2) (2019), 1800278.
- [50] L. Lu, et al., Cooperative plasmonic effect of Ag and Au nanoparticles on enhancing performance of polymer solar cells, *Nano Lett.* 13 (1) (2013) 59–64.
- [51] K. Huang, et al., Rapid laser annealing of silver electrodes for printing organic thin-film transistors on plastic substrates, *IEEE Trans. Electron. Dev.* 66 (6) (2019) 2729–2734.
- [52] A.P. Piedade, et al., Thin films for medical and environmental applications, in: *Thin Film Synthesis and Application for Medical and Biological Use*, IntechOpen, 2018.
- [53] G. Schmid, B. Corain, Nanoparticulated gold: syntheses, structures, electronics, and reactivities, *Eur. J. Inorg. Chem.* 2003 (17) (2003) 3081–3098.
- [54] M.G. Toudeshkhoei, et al., Microfluidic devices with gold thin film channels for chemical and biomedical applications: a review, *Biomed. Microdevices* 21 (4) (2019) 93.
- [55] S. De, R. Madhuri, Functionalized nanomaterials for electronics and electrical and energy industries, in: *Handbook of Functionalized Nanomaterials for Industrial Applications*, Elsevier, 2020, pp. 269–296.
- [56] D. Jariwala, et al., Carbon nanomaterials for electronics, optoelectronics, photovoltaics, and sensing, *Chem. Soc. Rev.* 42 (7) (2013) 2824–2860.
- [57] N.S. Alsaiani, et al., The application of nanomaterials for the electrochemical detection of antibiotics: a review, *Micromachines* 12 (3) (2021) 308.
- [58] M.I. Sohail, et al., Environmental application of nanomaterials: a promise to sustainable future, in: *Comprehensive Analytical Chemistry*, Elsevier, 2019, pp. 1–54.
- [59] H. Abdolmaleki, P. Kidmose, S. Agarwala, Droplet-based techniques for printing of functional inks for flexible physical sensors, *Adv. Mater.* 33 (20) (2021), 2006792.
- [60] E. Dimitriou, N.S. Michailidis, Printable conductive inks used for the fabrication of electronics: an overview, *Nanotechnology* 32 (50) (2021) 502009, <https://doi.org/10.1088/1361-6528/abeff1>. In this issue.
- [61] N. Turan, et al., Atmospheric pressure and ambient temperature plasma jet sintering of aerosol jet printed silver nanoparticles, *ACS Appl. Mater. Interfaces* 13 (39) (2021) 47244–47251.
- [62] L. Nayak, et al., A review on inkjet printing of nanoparticle inks for flexible electronics, *J. Mater. Chem. C* 7 (29) (2019) 8771–8795.
- [63] P. Calvert, Inkjet printing for materials and devices, *Chem. Mater.* 13 (10) (2001) 3299–3305.
- [64] C. Ru, et al., A review of non-contact micro- and nano-printing technologies, *J. Micromech. Microeng.* 24 (5) (2014), 053001.
- [65] S.Z. Hossain, et al., Development of a bioactive paper sensor for detection of neurotoxins using piezoelectric inkjet printing of sol–gel-derived bioinks, *Anal. Chem.* 81 (13) (2009) 5474–5483.
- [66] P.-H. Chen, et al., Pressure response and droplet ejection of a piezoelectric inkjet printhead, *Int. J. Mech. Sci.* 41 (2) (1999) 235–248.
- [67] T. Gomes, et al., Thermal inkjet printing of polyaniline on paper, *Thin Solid Films* 520 (24) (2012) 7200–7204.
- [68] X. Cui, et al., Thermal inkjet printing in tissue engineering and regenerative medicine, *Recent Pat. Drug Deliv. Formulation* 6 (2) (2012) 149–155.
- [69] E.B. Secor, Principles of aerosol jet printing, *Flex. Print. Electron.* 3 (3) (2018), 035002.
- [70] S.B. Balani, et al., Processes and materials used for direct writing technologies: a review, *Res. Eng.* 11 (2021), 100257.
- [71] N. Wilkinson, et al., A review of aerosol jet printing—a non-traditional hybrid process for micro-manufacturing, *Int. J. Adv. Manuf. Technol.* 105 (11) (2019) 4599–4619.
- [72] J. Park, et al., D.K. Mukhopadhyay, C.Y. Lee, M.S. Strano, A.G. Alleyne, J. G. Georgiadis, P.M. Ferreira, J.A. Rogers, *Nat. Mater.* 6 (782) (2007), 2100817.
- [73] J.-U. Park, et al., High-resolution electrohydrodynamic jet printing, *Nat. Mater.* 6 (10) (2007) 782–789.
- [74] M.S. Onses, et al., Mechanisms, capabilities, and applications of high-resolution electrohydrodynamic jet printing, *Small* 11 (34) (2015) 4237–4266.
- [75] J. George, Preparation of Thin Films, Taylor & Francis, 1992.
- [76] P. Martin, Ion-based methods for optical thin film deposition, *J. Mater. Sci.* 21 (1) (1986) 1–25.
- [77] M. Kemell, M. Ritala, M. Leskelä, Thin film deposition methods for CuInSe₂ solar cells, *Crit. Rev. Solid State Mater. Sci.* 30 (1) (2005) 1–31.
- [78] K. Seshan, Handbook of Thin Film Deposition Processes and Techniques, William Andrew, 2001.
- [79] D. Perednis, L.J. Gauckler, Thin film deposition using spray pyrolysis, *J. Electroceram.* 14 (2) (2005) 103–111.
- [80] J. Bourassa, et al., Water vapor-assisted sintering of silver nanoparticle inks for printed electronics, *SN Appl. Sci.* 1 (6) (2019) 1–6.
- [81] K. Rajan, et al., Silver nanoparticle ink technology: state of the art, *Nanotechnol. Sci. Appl.* 9 (2016) 1.
- [82] J. Niittynen, et al., Alternative sintering methods compared to conventional thermal sintering for inkjet printed silver nanoparticle ink, *Thin Solid Films* 556 (2014) 452–459.
- [83] Y.J. Moon, et al., Effect of thickness on surface morphology of silver nanoparticle layer during furnace sintering, *J. Electron. Mater.* 44 (4) (2015) 1192–1199.
- [84] D. Kim, J. Moon, Highly conductive ink jet printed films of nanosilver particles for printable electronics, *Electrochem. Solid State Lett.* 8 (11) (2005) J30.
- [85] I. Kim, et al., Selective light-induced patterning of carbon nanotube/silver nanoparticle composite to produce extremely flexible conductive electrodes, *ACS Appl. Mater. Interfaces* 9 (7) (2017) 6163–6170.
- [86] D.J. Lee, et al., Pulsed light sintering characteristics of inkjet-printed nanosilver films on a polymer substrate, *J. Micromech. Microeng.* 21 (12) (2011), 125023.
- [87] S. Wünsch, et al., Localized atmospheric plasma sintering of inkjet printed silver nanoparticles, *J. Mater. Chem.* 22 (47) (2012) 24569–24576.
- [88] S. Ma, et al., Low temperature plasma sintering of silver nanoparticles, *Appl. Surf. Sci.* 293 (2014) 207–215.
- [89] H.W. Tan, et al., Induction sintering of silver nanoparticle inks on polyimide substrates, *Adv. Mater. Technol.* 5 (1) (2020), 1900897.
- [90] Z.A. Munir, D.V. Quach, M. Ohyanagi, Electric current activation of sintering: a review of the pulsed electric current sintering process, *J. Am. Ceram. Soc.* 94 (1) (2011) 1–19.
- [91] S. Feng, et al., Maskless patterning of biodegradable conductors by selective laser sintering of microparticle inks and its application in flexible transient electronics, *ACS Appl. Mater. Interfaces* 11 (49) (2019) 45844–45852.
- [92] W. Liu, et al., Recent progress in rapid sintering of nanosilver for electronics applications, *Micromachines* 9 (7) (2018) 346.
- [93] N.K. Roy, et al., A comprehensive study of the sintering of copper nanoparticles using femtosecond, nanosecond, and continuous wave lasers, *J. Micro Nano-Manufacturing* 6 (1) (2018).
- [94] S.H. Ko, et al., Air stable high resolution organic transistors by selective laser sintering of ink-jet printed metal nanoparticles, *Appl. Phys. Lett.* 90 (14) (2007), 141103.
- [95] Q. Liu, et al., Picosecond laser sintering of silver paste printed by laser induced forward transfer, *Opt. Laser. Technol.* 135 (2021), 106712.
- [96] J. Yeo, et al., Flexible supercapacitor fabrication by room temperature rapid laser processing of roll-to-roll printed metal nanoparticle ink for wearable electronics application, *J. Power Sources* 246 (2014) 562–568.
- [97] S.H. Ko, et al., All-inkjet-printed flexible electronics fabrication on a polymer substrate by low-temperature high-resolution selective laser sintering of metal nanoparticles, *Nanotechnology* 18 (34) (2007), 345202.
- [98] M. Bram, et al., Application of electric current-assisted sintering techniques for the processing of advanced materials, *Adv. Eng. Mater.* 22 (6) (2020), 2000051.
- [99] E.G. Gamaly, A.V. Rode, Physics of ultra-short laser interaction with matter: from phonon excitation to ultimate transformations, *Prog. Quant. Electron.* 37 (5) (2013) 215–323.
- [100] M. Xiao, et al., Laser-induced joining of nanoscale materials: processing, properties, and applications, *Nano Today* 35 (2020), 100959.
- [101] B. Rethfeld, et al., Modelling ultrafast laser ablation, *J. Phys. Appl. Phys.* 50 (19) (2017), 193001.
- [102] L. Jiang, H.-L. Tsai, Improved two-temperature model and its application in ultrashort laser heating of metal films, 2005, pp. 1167–1173, 127(10).
- [103] R.K. Bordia, S.J.L. Kang, E.A. Olevsky, Current understanding and future research directions at the onset of the next century of sintering science and technology, *J. Am. Ceram. Soc.* 100 (6) (2017) 2314–2352.
- [104] S.-J.L. Kang, Sintering: Densification, Grain Growth and Microstructure, Elsevier, 2004.
- [105] D. Demirskyi, D. Agrawal, A. Ragulya, Neck growth kinetics during microwave sintering of copper, *Scripta Mater.* 62 (8) (2010) 552–555.
- [106] J. Frenkel, Viscous flow of crystalline bodies under the action of surface tension, *J. Phys.* 9 (1945) 385.
- [107] M.N. Rahaman, Ceramic Processing and Sintering, CRC press, 2017.
- [108] J. Yan, et al., Sintering mechanisms and mechanical properties of joints bonded using silver nanoparticles for electronic packaging applications, *Weld. World* 59 (3) (2015) 427–432.
- [109] G.C. Kuczynski, Self-diffusion in sintering of metallic particles, in: *Sintering Key Papers*, Springer, 1990, pp. 509–527.
- [110] R.L. Coble, Sintering crystalline solids. I. Intermediate and final state diffusion models, *J. Appl. Phys.* 32 (5) (1961) 787–792.
- [111] S.J. Kim, D.-J. Jang, Laser-induced nanowelding of gold nanoparticles, *Appl. Phys. Lett.* 86 (3) (2005), 033112.
- [112] H. Huang, et al., Controlled joining of Ag nanoparticles with femtosecond laser radiation, *J. Appl. Phys.* 112 (12) (2012), 123519.
- [113] Z. Jiao, et al., Formation and characterization of femtosecond-laser-induced subcluster segregated nanoalloys, *J. Phys. Chem. C* 118 (42) (2014) 24746–24751.
- [114] L.O. Herrmann, et al., Threading plasmonic nanoparticle strings with light, *Nat. Commun.* 5 (1) (2014) 1–6.
- [115] P. Peng, et al., Joining of silver nanomaterials at low temperatures: processes, properties, and applications, *ACS Appl. Mater. Interfaces* 7 (23) (2015) 12597–12618.
- [116] I. Theodorakos, et al., Selective laser sintering of Ag nanoparticles ink for applications in flexible electronics, *Appl. Surf. Sci.* 336 (2015) 157–162.
- [117] H. Huang, et al., Efficient localized heating of silver nanoparticles by low-fluence femtosecond laser pulses, *Appl. Surf. Sci.* 331 (2015) 392–398.
- [118] A. Anastasiou, et al., Exogenous mineralization of hard tissues using photo-absorptive minerals and femto-second lasers; the case of dental enamel, *Acta Biomater.* 71 (2018) 86–95.
- [119] S. Agarwala, et al., Wearable bandage-based strain sensor for home healthcare: combining 3D aerosol jet printing and laser sintering, *ACS Sens.* 4 (1) (2018) 218–226.

- [120] Y. Son, et al., Nanoscale electronics: digital fabrication by direct femtosecond laser processing of metal nanoparticles, *Adv. Mater.* 23 (28) (2011) 3176–3181.
- [121] J. Noh, J. Ha, D. Kim, Femtosecond and nanosecond laser sintering of silver nanoparticles on a flexible substrate, *Appl. Surf. Sci.* 511 (2020), 145574.
- [122] C.-W. Cheng, et al., Femtosecond laser melting of silver nanoparticles: comparison of model simulations and experimental results, *Appl. Phys. A* 124 (5) (2018) 1–8.
- [123] C.-W. Cheng, J. Chen, Femtosecond laser sintering of copper nanoparticles, *Appl. Phys. A* 122 (4) (2016) 1–8.
- [124] O. Ekici, et al., Thermal analysis of gold nanorods heated with femtosecond laser pulses, *J. Phys. D Appl. Phys.* 41 (18) (2008), 185501.
- [125] J. Guo, et al., Femtosecond Laser Sintering Al Nanoparticles: A Multiscale Investigation of Combined Molecular Dynamics Simulation and Two-Temperature Model, *Powder Technology*, 2022, 117682.
- [126] C. Symietz, et al., Femtosecond laser induced fixation of calcium alkali phosphate ceramics on titanium alloy bone implant material, *Acta Biomater.* 6 (8) (2010) 3318–3324.
- [127] E. Elmadani, et al., Characterization of rare-earth oxide photoactivated calcium phosphate minerals for resurfacing teeth, *J. Am. Ceram. Soc.* 95 (9) (2012) 2716–2724.
- [128] A. Anastasiou, et al., Sintering of calcium phosphates with a femtosecond pulsed laser for hard tissue engineering, *Mater. Des.* 101 (2016) 346–354.
- [129] R. Kurnoothala, K.C. Vishnubhatla, Facile fabrication of integrated microfluidic SERS substrate by femtosecond laser sintering of silver nano particles, *Opt. Mater.* 111 (2021), 110518.
- [130] M. Mizoshiri, et al., Effect of substrates on femtosecond laser pulse-induced reductive sintering of cobalt oxide nanoparticles, *Nanomaterials* 11 (12) (2021) 3356.
- [131] M. Mizoshiri, K. Yoshidomi, Cu patterning using femtosecond laser reductive sintering of CuO nanoparticles under inert gas injection, *Materials* 14 (12) (2021) 3285.
- [132] Y. Huang, et al., Copper circuits fabricated on flexible polymer substrates by a high repetition rate femtosecond laser-induced selective local reduction of copper oxide nanoparticles, *Opt Express* 29 (3) (2021) 4453–4463.
- [133] J.A. Creighton, D.G. Eadon, Ultraviolet–visible absorption spectra of the colloidal metallic elements, *J. Chem. Soc., Faraday Trans.* 87 (24) (1991) 3881–3891.
- [134] A. Serkov, et al., Laser sintering of gravure printed indium tin oxide films on polyethylene terephthalate for flexible electronics, *Sci. Rep.* 9 (1) (2019) 1–8.
- [135] D. Lee, et al., Non-vacuum, single-step conductive transparent ZnO patterning by ultra-short pulsed laser annealing of solution-deposited nanoparticles, *Appl. Phys. A* 107 (1) (2012) 161–171.
- [136] V. Volodin, et al., Picosecond infrared laser crystallization of Ge layers in Ge/Si multi-nanolayers for optoelectronic applications, in: *International Conference on Micro-and Nano-Electronics 2021*, SPIE, 2022.
- [137] N. Farid, et al., Improvement of electrical properties of ITO thin films by melt-free ultra-short laser crystallization, *J. Phys. Appl. Phys.* 54 (18) (2021), 185103.
- [138] N. Farid, et al., Femtosecond laser-induced crystallization of amorphous silicon thin films under a thin molybdenum layer, *ACS Appl. Mater. Interfaces* 13 (31) (2021) 37797–37808.
- [139] A. Sharif, et al., Non-melt selective enhancement of crystalline structure in molybdenum thin films using femtosecond laser pulses, *J. Phys. Appl. Phys.* 55 (11) (2021), 115301.
- [140] A. Sharif, et al., Femtosecond laser assisted crystallization of gold thin films, *Nanomaterials* 11 (5) (2021) 1186.
- [141] X.-P. Zhan, et al., Room temperature crystallization of amorphous silicon film by ultrashort femtosecond laser pulses, *Opt Laser. Technol.* 112 (2019) 363–367.
- [142] J.S. Hoppius, et al., Femtosecond laser crystallization of amorphous titanium oxide thin films, *Appl. Phys. Lett.* 113 (7) (2018), 071904.
- [143] S.-C. Chen, et al., Femtosecond laser crystallization for boosting the conversion efficiency of flexible ink-printing Cu (in, Ga) Se₂ thin film solar cells, in: *CLEO: Applications and Technology*, Optical Society of America, 2017.
- [144] G. Zhang, et al., Femtosecond laser-induced crystallization in amorphous Ge₂Sb₂Te₅ films, *Thin Solid Films* 474 (1–2) (2005) 169–172.
- [145] S. Katayama, et al., Femtosecond laser induced crystallization and permanent relief grating structures in amorphous inorganic (In 2 O 3+ 1 wt% TiO 2) films, *Appl. Phys. Lett.* 81 (5) (2002) 832–834.